Chapter 24

Campbell Biology, AP Edition, 12e

A cormorant spreads its small wings.

Chapter 24

The Origin of Species

Figure 24.1

This flightless cormorant is one of many species in the Galápagos Islands that are found nowhere else in the world. How did a bird that cannot fly come to live in this isolated place? When Darwin visited the Galápagos, he also was intrigued by its unique species, and later concluded that they must have originated in the islands from ancestors that had traveled from South America.

How do new species originate from existing species?

Over time, populations of a single species connected by gene flow can diverge genetically, giving rise to a new species. Begin with populations with gene flow. Three populations of a species are connected this way. A barrier to gene flow is established, such as a reproductive barrier (including different habitats or mating behaviors). One population begins to diverge genetically from the other two populations. Gene flow continues between the other two populations. Eventually, the lone population is reproductively isolated from the other two and a new species has formed.

Key Concepts

- **24.1The biological species concept emphasizes reproductive isolation****
- **24.2Speciation can take place with or without geographic separation****
- **24.3Hybrid zones reveal factors that cause reproductive isolation****

- **24.4Speciation can occur rapidly or slowly and can result from changes in few or many genes****

Advanced placement edition

Speciation may occur when two populations become reproductively isolated from one another **(BIG IDEA 1)**. This can lead to differences in timing and coordination of behavior and changes in modes of intercellular and intracellular signal transmission **(BIG IDEA 3)**. Reproductive isolation can also affect relationships with other species in the environment and bring about adaptation or morphological change **(BIG IDEA 4)**.

Study Tip

Make a table: Some processes that can lead to speciation only occur in allopatric (geographically separated) populations, while others also occur in sympatric (geographically overlapping) populations. To help you keep track of these processes and the geographic conditions under which they can occur, fill in the table as you read the chapter.

A partially filled table lists the process, and its occurrence in either an allopatric population or a sympatric population.

The table is as follows:

Process (include page number and figure number)	Can occur in allopatric populations (yes or no)	Can occur in sympatric populations (yes or no)
Sexual selection (Figure 24.12)	Yes	Yes
blank	blank	blank

When Darwin came to the Galápagos Islands, he noted that they were teeming with unique species, like the flightless bird in Figure 24.1. Later, he realized that these species had formed relatively recently, writing in his diary, “Both in space and time, we seem to be brought somewhat near to that great fact—that mystery of mysteries—the first appearance of new beings on this Earth.”

The “mystery of mysteries” that captivated Darwin is **speciation**, the process by which one species splits into two species. Speciation fascinated Darwin (and many biologists since) because it has produced the tremendous diversity of life, repeatedly yielding new species that differ from existing ones. Speciation also helps to explain the many features that organisms share (the unity of life): When one species splits into two, the species that result share many characteristics because they are descended from this common ancestor. For example, at the DNA sequence level, such similarities indicate that the flightless cormorant in Figure

24.1 is closely related to flying cormorants found in the Americas. This suggests that the flightless cormorant originated from an ancestral cormorant species that flew from the mainland to the Galápagos.

Speciation forms a conceptual bridge between ****__microevolution__, changes over time in allele frequencies in a population, and **macroevolution**, the broad pattern of evolution above the species level. An example of macroevolutionary change is the origin of new groups of organisms, such as mammals or flowering plants, through a series of speciation events. We examined microevolutionary mechanisms in Chapter 23, and we'll turn to macroevolution in Chapter 25. In this chapter, we'll explore the mechanisms by which new species originate from existing ones. First, let's establish what we actually mean by a "species."

Get Ready for This Chapter

In this chapter, you will learn about mechanisms that can lead to speciation, the process by which one species splits into two or more species. Before beginning this chapter, you should be able to do the following:

- Distinguish meiosis from mitosis (see Figure 13.10).
- Explain the chromosomal basis of inheritance (see Figure 15.4 and Concept 15.1).
- Describe how mutation, gene flow, genetic drift, and natural selection affect allele frequencies in populations (see Concept 23.3).

The following questions provide a quick check of your basic knowledge in these areas.

1. Which of the following statements about meiosis is correct?

```
     In meiosis, each cell division produces two
genetically identical daughter cells.
```

That's incorrect. Try again.

Incorrect. Review the following material: Meiosis in a diploid cell produces two haploid cells. In meiosis, the independent assortment of homologous chromosomes results in each daughter cell receiving one of 2n (where n is the haploid chromosome number) possible combinations of the maternal and paternal chromosomes. Crossing over may occur during meiosis, thereby producing recombinant chromosomes that carry genes derived from two different parents.

```
        Meiosis in a diploid cell produces diploid daughter
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```
        Meiosis produces cells that differ genetically from the
parent cell.
```

That's correct.

Correct! If you had trouble with this question, review the following material: Meiosis in a diploid cell produces two haploid cells that differ genetically from one another and from the diploid parent cell. This occurs in part because in meiosis, the independent assortment of homologous chromosomes results in each daughter cell receiving one of 2n (where n is the haploid chromosome number) possible combinations of the maternal and paternal chromosomes. Another reason is that crossing over may occur during meiosis, thereby producing recombinant chromosomes that carry genes derived from two different parents.

```
        Crossing over cannot occur during meiosis.
```

That's incorrect. Try again.

Incorrect. Review the following material: Meiosis in a diploid cell produces two haploid cells. In meiosis, the independent assortment of homologous chromosomes results in each daughter cell receiving one of 2n (where n is the haploid chromosome number) possible combinations of the maternal and paternal chromosomes. Crossing over may occur during meiosis, thereby producing recombinant chromosomes that carry genes derived from two different parents.

2. In a species that has two chromosomes, suppose that a male has genotype aa for a gene on chromosome 1 and genotype bb for a gene on chromosome 2. If a female of this species has genotype AA for the gene on chromosome 1 and genotype BB for the gene on chromosome 2, which of the following represents the possible genotypes in a cell of their offspring?

aB on chromosome 1, Ab on chromosome 2

That's incorrect. Try again.

Incorrect. Review the following material: Each sperm produced by the male contains one copy of chromosome 1 (carrying the a allele) and one copy of chromosome 2 (carrying the b allele), whereas each egg produced by the female contains one copy of chromosome 1 (carrying the A allele) and one copy of chromosome 2 (carrying the B allele).

AA on chromosome 1, bb on chromosome 2

That's incorrect. Try again.

Incorrect. Review the following material: Each sperm produced by the male contains one copy of chromosome 1 (carrying the a allele) and one copy of chromosome 2 (carrying the b allele), whereas each egg produced by the female contains one copy of chromosome 1 (carrying the A allele) and one copy of chromosome 2 (carrying the B allele).

AB on chromosome 1, ab on chromosome 2

That's incorrect. Try again.

Incorrect. Review the following material: Each sperm produced by the male contains one copy of chromosome 1 (carrying the a allele) and one copy of chromosome 2 (carrying the b allele), whereas each egg produced by the female contains one copy of chromosome 1 (carrying the A allele) and one copy of chromosome 2 (carrying the B allele).

Aa on chromosome 1, Bb on chromosome 2

That's correct.

Correct! If you had trouble with this question, review the following material: Each sperm produced by the male contains one copy of chromosome 1 (carrying the a allele) and one copy of chromosome 2 (carrying the b allele), whereas each egg produced by the female contains one copy of chromosome 1 (carrying the A allele) and one copy of chromosome 2 (carrying the B allele). Thus, when any egg and sperm cells unite, the offspring cells will have genotype Aa on chromosome 1 and genotype Bb on chromosome 2.

3. A windstorm transports a few seeds of a plant species from Island A, where the species is abundant, to Island B, where prior to this event the plant species was not found. The environmental conditions on the two islands are very similar, but the islands are located far away from one another. Over time, a small population of this plant becomes established on Island B. A biologist samples plants on both islands and finds that allele frequencies on Island A differ substantially from allele frequencies on Island B at several genetic loci not known to affect survival or reproduction. Based on this information, the observed genetic changes mostly likely resulted from

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genetic drift.
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That's correct.

Correct! If you had trouble with this question, review the following material: Since, the environmental conditions on the two islands are very similar and the genetic loci sampled are not known to affect survival or reproduction, it is not likely that the observed differences were caused by natural selection. Gene flow causes populations to become more similar genetically over time, not more different. However, the observed differences could well be caused by genetic drift, the fluctuation of allele frequencies from one generation to the next in a population due to chance events; for example, the founder effect often causes members of a new population (such as that on Island B) to differ genetically from members of the source population (Island A).

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natural selection.
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That's incorrect. Try again.

Incorrect. Review the following material: Since the environmental conditions on the two islands are very similar and the genetic loci sampled are not known to affect survival or reproduction, it is not likely that the observed differences were caused by natural selection. Gene flow causes populations to become more similar genetically over time, not more different.

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mutation.
```

That's incorrect. Try again.

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gene flow (after the species became established on Island B).

That's incorrect. Try again.

Incorrect. Review the following material: Since the environmental conditions on the two islands are very similar and the genetic loci sampled are not known to affect survival or reproduction, it is not likely that the observed differences were caused by natural selection. Gene flow causes populations to become more similar genetically over time, not more different.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 24.1Define the biological species concept, and identify reproductive barriers that could isolate a species' gene pool from that of other species.
- 24.2Compare and contrast allopatric speciation and sympatric speciation.
- 24.3Explain what a hybrid zone is, and identify possible outcomes for a hybrid zone over time.
- 24.4Give examples that show how speciation can occur rapidly or slowly, and can result from changes in few or many genes.

Concept 24.1: The biological species concept emphasizes reproductive isolation

The word *species* is Latin for “kind” or “appearance.” In daily life, we commonly distinguish between various “kinds” of organisms—dogs and cats, for instance—based on differences in their appearance. But are organisms truly divided into the discrete units we call species? To answer this question, biologists compare not only the morphology (body form) of different groups of organisms but also their physiology, biochemistry, and DNA sequences. The results generally confirm that morphologically distinct species are indeed discrete groups, differing in many ways besides their body forms.

The Biological Species Concept

The primary definition of species used in this textbook is the **biological species concept**. According to this concept, a **species** is a group of populations whose members have the potential to interbreed in nature and produce viable, fertile offspring—but do not produce viable, fertile offspring with members of other such groups **(Figure 24.2)**. Thus, the members of a biological species are united by being reproductively compatible, at least potentially. All human beings, for example, belong to the same species. A man and a woman from opposite sides of the globe may be unlikely to meet and mate, but if they do, they could have viable babies who develop into fertile adults. In contrast, humans and chimpanzees remain distinct biological species, even where they live in the same region, because many factors keep them from interbreeding and producing fertile offspring.

Figure 24.2

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    The biological species concept is based on the potential to
interbreed, not on physical similarity.
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The biological species concept is based on the potential to interbreed, not on physical similarity.

Part a. Similarity between different species. The eastern meadowlark (Sturnella magna, left) and the western meadowlark (Sturnella neglecta, right) have similar body shapes and colorations. Nevertheless, they are distinct biological species because their songs and other behaviors are different enough to prevent interbreeding should they meet in the wild.

Part b. Diversity within a species. As diverse as we may be in appearance, all humans belong to a single biological species (Homo sapiens), defined by our capacity to interbreed successfully. Six images show people from different ethnic groups with different features such as hair, eye, and skin color.

What holds the gene pool of a species together, causing its members to resemble each other more than they resemble members of other species? Recall the evolutionary mechanism of *gene flow*, the transfer of alleles between populations (see Concept 23.3). Typically, gene flow occurs between the different populations of a species. This ongoing exchange of alleles tends to hold the populations together genetically. But as we'll explore in this chapter, a reduction or lack of gene flow can play a key role in the formation of new species.

Reproductive Isolation

Because biological species are defined in terms of reproductive compatibility, the formation of a new species hinges on **reproductive isolation**—the existence of biological factors (barriers) that impede members of two species from interbreeding and producing viable, fertile offspring **(Figure 24.3)**. Such barriers block gene flow between the species and limit the formation of **hybrids**, offspring that result from an interspecific mating. Although a single barrier may not prevent all gene flow, a combination of several barriers can effectively isolate a species' gene pool.

Figure 24.3

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    Exploring Reproductive Barriers
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Exploring reproductive barriers. Prezygotic barriers impede mating or hinder fertilization if mating does occur. They include habitat isolation, temporal isolation, behavioral isolation, mechanical isolation, and gametic isolation. Exploring reproductive barriers. Prezygotic barriers include gametic isolation. Postzygotic barriers include reduced hybrid viability, reduced hybrid fertility, and hybrid breakdown.

Prezygotic barriers include habitat isolation, temporal isolation, behavioral isolation, mechanical isolation, and gametic isolation, each described as follows.

- Habitat isolation: Two species that occupy different habitats within the same area may encounter each other rarely, if at all, even though they are not isolated by obvious physical barriers, such as mountain ranges. Example: These two fly species in the genus *Rhagoletis* occur in the same geographic areas, but the apple maggot fly (*Rhagoletis pomonella*) feeds and mates on hawthorns and apples while its close relative, the blueberry maggot fly (*R mendax*), mates and lays its eggs only on blueberries.
- Temporal isolation: Species that breed during different times of the day, different seasons, or different years cannot mix their gametes. Example: In North America, the geographic ranges of the western spotted skunk (*Spilogale gracilis*) and the eastern spotted skunk (*Spilogale putorius*) overlap, but *S. gracilis* mates in late summer and *S. putorius* mates in late winter.
- Behavioral isolation: Courtship rituals that attract mates and other behaviors unique to a species are effective reproductive barriers, even between closely related species. Such behavioral rituals enable mate recognition—a way to identify potential mates of the same species. Example: Blue-footed boobies, inhabitants of the Galápagos, mate only after a courtship display unique to their species. Part of the script calls for the male to high-step, a behavior that calls the female's attention to his bright blue feet.
- Mechanical isolation: Mating is attempted, but morphological differences prevent its successful completion. Example: The shells of two species of snails in the genus *Bradybaena* spiral in different directions: Moving inward to the center, one spirals in a counterclockwise direction, the other in a clockwise direction. As a result, the snails' genital openings (indicated by arrows) are not aligned, and mating cannot be completed.

Gametic isolation: Sperm of one species may not be able to fertilize the eggs of another species. For instance, sperm may not be able to survive in the reproductive tract of females of the other species, or biochemical mechanisms may prevent the sperm from penetrating the membrane surrounding the other species' eggs. Example: Gametic isolation separates certain closely related species of aquatic animals, such as sea urchins. Sea urchins release their sperm and eggs into the surrounding water, where they fuse and form zygotes. It is difficult for gametes of different species, such as the red and purple urchins shown here, to fuse because proteins on the surfaces of the eggs and sperm bind very poorly to each other.

Postzygotic barriers include reduced hybrid viability, reduced hybrid fertility, and hybrid breakdown, each described as follows.

- Reduced hybrid viability: The genes of different parent species may interact in ways that impair the hybrid's development or survival in its environment. Example: Some salamander subspecies of the genus Ensatina live in the same regions and habitats, where they may occasionally hybridize. But most of the hybrids do not complete development, and those that do are frail.
- Reduced hybrid fertility: Even if hybrids are vigorous, they may be sterile. If the chromosomes of the two parent species differ in number or structure, meiosis in the hybrids may fail to produce normal gametes. Since the infertile hybrids cannot produce offspring when they mate with either parent species, genes cannot flow freely between the species. Example: The hybrid offspring of a male donkey and a female horse is a mule, which is robust but sterile. A hinny, the offspring of a female donkey and a male horse, is also sterile.
- Hybrid breakdown: Some first-generation hybrids are viable and fertile, but when they mate with one another or with either parent species, offspring of the next generation are feeble or sterile. Example: Strains of cultivated rice have accumulated different mutant recessive alleles at two loci in the course of their divergence from a common ancestor. Hybrids between them are vigorous and fertile, but plants in the next generation that carry too many of these recessive alleles are small and sterile. Although these rice strains are not yet considered different species, they have begun to be separated by postzygotic barriers.

Video

Blue-footed Boobies Courtship Ritual

image

Play

image

Play

Clearly, a fly cannot mate with a frog or a fern, but the reproductive barriers between more closely related species are not so obvious. As described in Figure 24.3, these barriers can be classified according to whether they contribute to reproductive isolation before or after fertilization. **Prezygotic barriers** ("before the zygote") block fertilization from occurring. Such barriers typically act in one of three ways: by impeding members of different species from attempting to mate, by preventing an attempted mating from being completed successfully, or by hindering fertilization if mating is completed successfully. If a sperm cell from

one species overcomes prezygotic barriers and fertilizes an ovum from another species, a variety of **postzygotic barriers** ("after the zygote") may contribute to reproductive isolation after the hybrid zygote is formed. Developmental errors may reduce survival among hybrid embryos. Or problems after birth may cause hybrids to be infertile or decrease their chance of surviving long enough to reproduce.

Video

Prezygotic Barriers to Mating in Galápagos Finches by Peter and Rosemary Grant

image

Play

image

Play

Limitations of the Biological Species Concept

One strength of the biological species concept is that it directs our attention to a way by which speciation can occur: by the evolution of reproductive isolation. However, the number of species to which this concept can be usefully applied is limited. There is, for example, no way to evaluate the reproductive isolation of fossils. The biological species concept also does not apply to organisms that reproduce asexually all or most of the time, such as prokaryotes. (Many prokaryotes do transfer genes among themselves, as we will discuss in Concept 27.2, but this is not part of their reproductive process.) Furthermore, in the biological species concept, species are designated by the *absence* of gene flow. But there are many pairs of species that are morphologically and ecologically distinct, and yet gene flow occurs between them. An example is the grizzly bear (*Ursus arctos*) and polar bear (*Ursus maritimus*), whose hybrid offspring have been dubbed "grolar bears" **(Figure 24.4)**. As we'll discuss, natural selection can cause such species to remain distinct even though some gene flow occurs between them. Because of the limitations to the biological species concept, alternative species concepts are useful in certain situations.

Figure 24.4

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    Hybridization between two species of bears in the genus
*Ursus*.
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Three bears: a, Grizzly bear (U arctos); b, Polar bear (U maritimus); c, Hybrid grolar bear.

Other Definitions of Species

While the biological species concept emphasizes the *separateness* of different species due to reproductive barriers, several other definitions emphasize the *unity within* a species. For example, the **morphological species concept** distinguishes a species by body shape and other structural features. The morphological species concept can be applied to asexual and sexual organisms, and it can be useful even without information on the extent of gene flow. In practice, scientists often distinguish species using morphological criteria. A disadvantage of this approach, however, is that it relies on subjective criteria; researchers may disagree on which structural features distinguish a species.

The **ecological species concept** defines a species in terms of its ecological niche, the sum of how members of the species interact with the nonliving and living parts of their environment (see Concept 54.1). For example, two species of oak trees might differ in their size or in their ability to tolerate dry conditions, yet still occasionally interbreed. Because they occupy different ecological niches, these oaks would be considered separate species even though they are connected by some gene flow. Unlike the biological species concept, the ecological species concept can accommodate asexual as well as sexual species. It also emphasizes the role of disruptive natural selection as organisms adapt to different environments.

In addition to those discussed here, more than 20 other species definitions have been proposed. The usefulness of each definition depends on the situation and the research questions being asked. For our purposes of studying how species originate, the biological species concept, with its focus on reproductive barriers, is particularly helpful.

HHMI BioInteractive Video

The Origin of Species: The Beak of the Finch

image

Play

image

Play

For more resources from HHMI BioInteractive, go to www.biointeractive.org.

Concept 24.2: Speciation can take place with or without geographic separation

Having discussed what constitutes a unique species, let's return to the process by which such species arise from existing species. We'll describe this process by focusing on the geographic setting in which gene flow is interrupted between populations of the existing species—in allopatric speciation the populations are geographically isolated, while in sympatric speciation they are not **(Figure 24.5)**.

Figure 24.5

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The geography of speciation.
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Gold fish in a pond with a barrier partially submerged. If the pools are divided, allopatric speciation can occur. If the pools remain unchanged sympatric speciation can occur.

Gold fish are in a pond with a barrier in the middle that is partially submerged. If the water level drops, the pools are divided and allopatric speciation can occur which is shown as two blue fish in the left pool and five gold fish in the right pool. If the water level remains unchanged the fish are not divided and sympatric speciation can occur which is shown as two blue fish among eight gold fish.

- a. Allopatric speciation: A population forms a new species while geographically isolated from its parent population.
- b. Sympatric speciation: A subset of a population forms a new species without geographic separation.

Allopatric ("Other Country") Speciation

In **allopatric speciation** (from the Greek *allos*, other, and *patra*, homeland), gene flow is interrupted when a population is divided into geographically isolated subpopulations. For example, the water level in a lake may subside, resulting in two or more smaller lakes that are now home to separated populations (see Figure 24.5a). Or a river may change course and divide a population of animals that cannot cross it. Allopatric speciation can also occur without geologic change, such as when individuals colonize a remote area and their descendants become geographically isolated from the parent population. The flightless

cormorant shown in Figure 24.1 probably originated in this way from an ancestral flying species that reached the Galápagos Islands.

The Process of Allopatric Speciation

How formidable must a geographic barrier be to promote allopatric speciation? The answer depends on the ability of the organisms to move about. Birds, mountain lions, and coyotes can cross rivers and canyons—as can windblown pollen and the seeds of some flowering plants. In contrast, small rodents may find a river or canyon a formidable barrier.

Once geographic isolation has occurred, the separated gene pools may diverge. Different mutations arise, and natural selection and genetic drift may alter allele frequencies in different ways in the separated populations. Reproductive isolation may then evolve as a by-product of the genetic divergence that results from selection or drift.

Figure 24.6 describes an example. On Andros Island, in the Bahamas, populations of the mosquitofish *Gambusia hubbsi* colonized a series of ponds that later became isolated from one another. Genetic analyses indicate that little or no gene flow currently occurs between the ponds. The environments of these ponds are very similar except that some contain predatory fishes, while others do not. In ponds with predatory fishes, selection has favored the evolution of a mosquitofish body shape that enables rapid bursts of speed (see Figure 24.6a). In ponds without predatory fishes, selection has favored a different body shape, one that improves the ability to swim for long periods of time (see Figure 24.6b).

Figure 24.6

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Evolution in mosquitofish populations.
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Evolution in mosquitofish populations.

Different body shapes have evolved in mosquitofish populations from ponds with and without predators. These differences affect how quickly the fishes can accelerate to escape and their survival rate when exposed to predators.

- Part a. Differences in body shape: High predation can lead to differences in body shape that enable quick getaways, for example in mosquitofish. In ponds with predatory fishes, the head of the mosquitofish is streamlined and the tail is powerful, enabling rapid bursts of speed. Under low predation. In ponds without predatory fishes, mosquitofish have a different body shape that favors long, steady swimming.

- Part b. Differences in escape acceleration and survival: mosquitofish in ponds containing predators show faster escape acceleration and higher survival rates than mosquitofish in ponds without predators.

Two bar graphs. Y-axis of graph on left is labeled escape acceleration (meters per s squared) and is marked from 50 to 110 in increments of 10. Mosquitofish from source ponds with predators present show an escape acceleration of almost 100 meters per s squared (orange bar). Mosquitofish from ponds with no predators show a much lower escape acceleration of approximately 70 meters per s squared (green bar). Y-axis of graph on right is labeled survival rate with predators and is marked from zero to 1.0 in increments of 0.2. Mosquitofish from ponds with predators present have a higher survival rate than those from ponds with no predators: 0.9 (orange bar) compared to 0.4 (green bar).

How have these different selective pressures affected the evolution of reproductive barriers? Researchers studied this question by bringing together mosquitofish from the two types of ponds. They found that female mosquitofish prefer to mate with males whose body shape is similar to their own. This preference establishes a behavioral barrier to reproduction between mosquitofish from ponds with predators and those from ponds without predators. Thus, as a by-product of selection for avoiding predators, reproductive barriers have formed in these allopatric populations.

Evidence of Allopatric Speciation

Many studies provide evidence that speciation can occur in allopatric populations. For example, laboratory studies show that reproductive barriers can develop when populations are isolated experimentally and subjected to different environmental conditions **(Figure 24.7)**.

Figure 24.7

Inquiry: Can divergence of allopatric populations lead to reproductive isolation?

Experiment

A researcher divided a laboratory population of the fruit fly *Drosophila pseudoobscura*, raising some flies on a starch medium and others on a maltose medium. After one year (about 40 generations), natural selection resulted in divergent evolution: Populations raised on starch digested starch more efficiently, while those

raised on maltose digested maltose more efficiently. The researcher then put flies from the same or different populations in mating cages and measured mating frequencies. All flies used in the mating preference tests were reared for one generation on a standard cornmeal medium.

Can divergence of allopatric populations lead to reproductive isolation?

Experiment. A researcher divided a laboratory population of the fruit fly Drosophila pseudoobscura, raising some flies on a starch medium and others on a maltose medium. After one year (about 40 generations), natural selection resulted in divergent evolution: Populations raised on starch digested starch more efficiently, while those raised on maltose digested maltose more efficiently. The researcher then put flies from the same or different populations in mating cages and measured mating frequencies. All flies used in the mating preference tests were reared for one generation on a standard cornmeal medium.

Results. Mating patterns among populations of flies raised on different media are shown below. When flies from starch populations were mixed with flies from maltose populations, the flies tended to mate with like partners (see the first table below). But in a control group consisting of flies from two different populations adapted to starch, the flies were about as likely to mate with flies of the other population as with flies of their own population (see the second table below); similar results were obtained for control groups adapted to maltose.

Results

Mating patterns among populations of flies raised on different media are shown below. When flies from “starch populations” were mixed with flies from “maltose populations,” the flies tended to mate with like partners (see the lefthand table). But in a control group consisting of flies from two different populations adapted to starch, the flies were about as likely to mate with flies of the other population as with flies of their own population (see the righthand table); similar results were obtained for control groups adapted to maltose.

Results of an experiment on Can divergence of allopatric populations lead to reproductive isolation?

	Female
Starch	Maltose
Male	Starch
Maltose	8
Number of matings in experimental group	

	Female
Starch population 1	Starch population 2
Male	Starch population 1
Starch population 2	12
Number of matings in control group	

Conclusion

In the experimental group, the strong preference of “starch flies” and “maltose flies” to mate with like-adapted flies indicates that a reproductive barrier was forming between these fly populations. Although this barrier was not absolute (some mating between starch flies and maltose flies did occur), after 40 generations reproductive isolation appeared to be increasing. This barrier may have been caused by differences in courtship behavior that arose as an incidental by-product of differing selective pressures as these allopatric populations adapted to different sources of food.

Data from D. M. B. Dodd, Reproductive isolation as a consequence of adaptive divergence in *Drosophila pseudoobscura, Evolution* 43:1308–1311 (1989).

WHAT IF?***Why were all flies used in the mating preference tests reared on a standard medium (rather than on starch or maltose)?*

*For suggested answer, see*Appendix A.**

Field studies indicate that allopatric speciation also can occur in nature. Consider the 30 species of snapping shrimp in the genus *Alpheus* that live off the Isthmus of Panama, the land bridge that connects South and North America **(Figure 24.8)**. Fifteen of these species live on the Atlantic side of the isthmus, while the other 15 live on the Pacific side. Before the isthmus formed, gene flow could occur between the Atlantic and Pacific populations of snapping shrimp. Did the species on different sides of the isthmus originate by allopatric speciation? Morphological and genetic data group these shrimp into 15 pairs of *sister species*, pairs whose member species are each other’s closest relative. In each of these 15 pairs, one of the sister species lives on the Atlantic side of the isthmus, while the other lives on the Pacific side. This fact strongly suggests that the two species arose as a consequence of geographic separation. Furthermore, genetic analyses indicate that the *Alpheus* species originated from 9 to 3 million years ago, with the sister species that live in the deepest water diverging first. These divergence times are consistent with geologic evidence that the isthmus formed gradually, starting 10 million years ago, and closing completely about 3 million years ago.

Figure 24.8

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Allopatric speciation in snapping shrimp (*Alpheus*).
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A map of the Isthmus of Panama is shown with A formosus and A nuttingi shrimp in the Atlantic Ocean, and A panamensis (sister A formosus) and A millsae (sister A nuttingi) in the Pacific Ocean.

The shrimp pictured are just 2 of the 15 pairs of sister species that arose as populations were divided by the formation of the Isthmus of Panama. The color-coded type indicates the sister species.

The importance of allopatric speciation is also suggested by the fact that regions that are isolated or highly subdivided by barriers typically have more species than do otherwise similar regions that lack such features. For example, many unique plants and animals are found on the geographically isolated Hawaiian Islands (we'll return to the origin of Hawaiian species in Concept 25.4). Field studies also show that reproductive isolation between two populations generally increases as the geographic distance between them increases, a finding consistent with allopatric speciation. In the **Scientific Skills Exercise**, you can analyze data from one such study that examined reproductive isolation in geographically separated salamander populations.

Scientific Skills Exercise

Identifying Independent and Dependent Variables, Making a Scatter Plot, and Interpreting Data

AP Science Practices 2, 3, 4, 5

Does Distance Between Salamander Populations Increase Their Reproductive Isolation? Allopatric speciation begins when populations become geographically isolated, reducing mating between individuals in different populations and thus reducing gene flow. It is logical that as distance between populations increases, so will their degree of reproductive isolation. To test this hypothesis, researchers studied populations of the dusky salamander (*Desmognathus ochrophaeus*) living on different mountain ranges in the southern Appalachians.

A salamander.

How the Experiment Was Done The researchers tested the reproductive isolation of pairs of salamander populations by leaving one male and one female together and later checking the females for the presence of

sperm. Four mating combinations were tested for each pair of populations (A and B)—two *within* the same population (female A with male A and female B with male B) and two *between* populations (female A with male B and female B with male A).

Data from the Experiment The researchers used an index of reproductive isolation that ranged from a value of 0 (no isolation) to a value of 2 (full isolation). The proportion of successful matings for each mating combination was measured, with 100% success=1 and no success=0. The reproductive isolation value for two populations is the sum of the proportion of successful matings of each type within populations (AA+BB) minus the sum of the proportion of successful matings of each type between populations (AB+BA). The table provides distance and reproductive isolation data for 27 pairs of dusky salamander populations.

A table lists geographic distance and reproductive isolation values.

Geographic distance, kilometers	**15**	**32**	**40**	**47**	**42**	**62**	**63**	**81**	**86**	**107**	**107**
Reproductive isolation value	0.32	0.54	0.5	0.5	0.82	0.37	0.67	0.53	1.15	0.73	0.82
Distance (continued)	137	150	165	189	219	239	247	53	55	62	105
Isolation (continued)	0.5	0.57	0.91	0.93	1.5	1.22	0.82	0.99	0.21	0.56	0.41

Data from S. G. Tilley, A. Verrell, and S. J. Arnold, Correspondence between sexual isolation and allozyme differentiation: a test in the salamander *Desmognathus ochrophaeus, Proceedings of the National Academy of Sciences USA* 87:2715–2719 (1990).

INTERPRET THE DATA

- State the researchers' hypothesis, and identify the independent and dependent variables in this study. Explain why the researchers used four mating combinations for each pair of populations.
- Calculate the value of the reproductive isolation index
- if *all* of the matings within a population were successful, but *none* of the matings between populations were successful;
- if salamanders are equally successful in mating with members of their own population and members of another population.

- Make a scatter plot to help you visualize any patterns that might indicate a relationship between the variables. Plot the independent variable on the x-axis and the dependent variable on the y-axis. (For additional information about graphs, see the Scientific Skills Review in Appendix D.)
- Interpret your graph by
- explaining in words any pattern indicating a possible relationship between the variables and
- hypothesizing the possible cause of such a relationship.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Note that while geographic isolation prevents interbreeding between members of allopatric populations, physical separation is not a biological barrier to reproduction. Biological reproductive barriers such as those described in Figure 24.3 are intrinsic to the organisms themselves. Hence, it is biological barriers that can prevent interbreeding when members of different populations come into contact with one another.

Sympatric ("Same Country") Speciation

In **sympatric speciation** (from the Greek *syn*, together), speciation occurs in populations that live in the same geographic area (see Figure 24.5b). How can reproductive barriers form between sympatric populations while their members remain in contact with each other? Although such contact (and the ongoing gene flow that results) makes sympatric speciation less common than allopatric speciation, sympatric speciation can occur if gene flow is reduced by such factors as polyploidy, sexual selection, and habitat differentiation. (Note that sexual selection and habitat differentiation also can promote allopatric speciation.)

Polyploidy

A species may originate from an accident during cell division that results in extra sets of chromosomes, a condition called **polyploidy**. Polyploid speciation occasionally occurs in animals; for example, the gray tree frog *Hyla versicolor* (see Figure 23.16) is thought to have originated in this way. However, polyploidy is far more common in plants. In fact, botanists estimate that more than 80% of the plant species alive today are descended from ancestors that formed by polyploid speciation.

Two distinct forms of polyploidy have been observed in plant (and a few animal) populations. An **autopolyploid** (from the Greek *autos*, self) is an individual that has more than two chromosome sets that are all derived from a single species. In plants, for example, a failure of cell division could double a cell's chromosome number from the original number (2n) to a tetraploid number (4n) **(Figure 24.9)**.

Figure 24.9

```
Sympatric speciation by autopolyploidy.
```

Autopolyploid speciation. Cell division error leads to new species (4 n).

Cell is 2 n = 6 (3 sets of chromosomes). After cell division error, tetraploid cell results (4 n): there are four of each chromosome instead of two, for 12 total chromosomes. Tetraploid cell undergoes meiosis and the gametes are 2 n = 6. After fertilization, a new 4 n species results with 12 chromosomes.

A tetraploid can produce fertile tetraploid offspring by self-pollinating or by mating with other tetraploids. In addition, the tetraploids are reproductively isolated from 2n plants of the original population because the triploid (3n) offspring of such unions have reduced fertility. Thus, in just one generation, autopolyploidy can generate reproductive isolation without any geographic separation.

A second form of polyploidy can occur when two different species interbreed and produce hybrid offspring. Most such hybrids are sterile because the set of chromosomes from one species cannot pair during meiosis with the set of chromosomes from the other species. However, an infertile hybrid may be able to propagate itself asexually (as many plants can do). In subsequent generations, various mechanisms can change a sterile hybrid into a fertile polyploid called an **allopolyploid(Figure 24.10)**. The allopolyploids are fertile when mating with each other but cannot interbreed with either parent species; thus, they represent a new biological species. The new species has a diploid chromosome number equal to the sum of the diploid chromosome numbers of the two parent species.

Figure 24.10

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One mechanism for allopolyploid speciation in plants.
```

Gametes from Species A and Species B form a sterile hybrid with 5 chromosomes. Mitotic or meiotic error doubles the chromosome number and a new viable, fertile species is formed.

Most hybrids are sterile because their chromosomes are not homologous and cannot pair during meiosis. However, such a hybrid may be able to reproduce asexually. This diagram traces one mechanism by which sterile hybrids can give rise to fertile allopolyploids that are members of a new species.

Species A (2 n = 6) shown with a normal gamete (n = 3) along with Species B (2 n = 4) and normal gamete (n = 2). A sterile hybrid is formed with 5 chromosomes. Mitotic or meiotic error doubles the chromosome number and a new species is formed. The new species is a viable, fertile hybrid (allopolyploid) with 2 n = 10.

Figure 24.10 Walkthrough

One mechanism for allopolyploid speciation in plants

Watch an explanation of this figure in the Figure 24.10 Walkthrough: One mechanism for allopolyploid speciation in plants.

image

Play

image

Play

Although it can be challenging to study speciation in the field, scientists have documented at least five new plant species that have originated by polyploid speciation since 1850. One of these examples involves the origin of a new species of goatsbeard plant (genus *Tragopogon*) in the Pacific Northwest. *Tragopogon* first arrived in the region when humans introduced three European species in the early 1900s: *T. pratensis, T. dubius*, and *T. porrifolius*. These three species are now common in the Pacific Northwest and many other parts of North America. In 1950, a new *Tragopogon* species was discovered near the Idaho-Washington border, a region where all three European species also were found. Genetic analyses revealed that this new species, *Tragopogon miscellus*, is an allopolyploid that originated from a hybrid of two of the European species **(Figure 24.11)**. Although the *T. miscellus* population grows mainly by reproduction of its own members, hybridization between the parent species (followed by a mitotic or meiotic error that doubles the chromosome number) continues to add new members to the *T. miscellus* population. Later, scientists discovered another new *Tragopogon* species, *T. mirus*, which also arose by polyploid speciation.

Figure 24.11

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Allopolyploid speciation in *Tragopogon*.
```

Parent species T dubius (12) and T pratensis (12) form hybrid species T miscellus (24). Parent T dubius (12) and T porrifolius (12) form hybrid T mirus (24).

The gray boxes indicate the parent species of the two new polyploid species. The diploid chromosome number of each species is shown in parentheses.

Parent species T dubius (12) and T pratensis (12), both with yellow flowers, form hybrid species T miscellus (24) which also has yellow flowers. Parent T dubius (12) with yellow flowers and T porrifolius (12) with purple flowers form hybrid T mirus (24) with pink flowers.

VISUAL SKILLS *Based on this diagram, identify the two parent species of each polyploid species.*

*For suggested answer, see*Appendix A.**

Many important agricultural crops—such as oats, cotton, potatoes, tobacco, and wheat—are polyploids. For example, the wheat used for bread, *Triticum aestivum*, is an allohexaploid (six sets of chromosomes, two sets from each of three different species). The events that eventually led to modern wheat probably began about 8,000 years ago in the Middle East when an early cultivated wheat species hybridized with a wild grass. Today, plant geneticists generate new polyploids in the laboratory by using chemicals that induce meiotic and mitotic errors. By harnessing the evolutionary process, researchers can produce new polyploid species with desired qualities, such as a polyploid that combines the high yield of wheat with the hardiness of rye.

Animation

Speciation by Changes in Ploidy

image

Play

image

Sexual Selection

There is evidence that sympatric speciation can also be driven by sexual selection. Clues to how this can occur have been found in cichlid fish from one of Earth's hot spots of animal speciation, East Africa's Lake Victoria. This lake was once home to as many as 600 species of cichlids. Genetic data indicate that these species originated within the last 100,000 years from a small number of colonizing species that arrived from other lakes and rivers. How did so many species—more than double the number of freshwater fish species known in all of Europe—originate within a single lake?

One hypothesis is that subgroups of the original cichlid populations adapted to different food sources and the resulting genetic divergence contributed to speciation in Lake Victoria. But sexual selection, in which (typically) females select males based on their appearance (see Concept 23.4), may also have been a factor. Researchers have studied two closely related sympatric species of cichlids that differ mainly in the coloration of breeding males: Breeding *Pundamilia pundamilia* males have a blue-tinged back, whereas breeding *Pundamilia nyererei* males have a red-tinged back **(Figure 24.12)**. Their results suggest that mate choice based on male breeding coloration can act as a reproductive barrier that keeps the gene pools of these two species separate.

Figure 24.12

Inquiry: Does sexual selection in cichlids result in reproductive isolation?

Experiment

Researchers placed males and females of *Pundamilia pundamilia* and *P. nyererei* together in two aquarium tanks, one with natural light and one with a monochromatic orange lamp. Under normal light, the two species are noticeably different in male breeding coloration; under monochromatic orange light, the two species are very similar in color. The researchers then observed the mate choices of the females in each tank.

Does sexual selection in cichlids result in reproductive isolation?

Experiment. Researchers placed males and females of Pundamilia pundamilia and P nyererei together in two aquarium tanks, one with natural light and one with a monochromatic orange lamp. Under normal light, the two species are noticeably different in male breeding coloration; under monochromatic orange light, the two species are very similar in color. The researchers then observed the mate choices of the females in each tank. Results. Under normal light, females of each species strongly preferred males of their own species. But under orange light, females of each species responded indiscriminately to males of both species. The resulting hybrids were viable and fertile.

Conclusion. The researchers concluded that mate choice by females based on male breeding coloration can act as a reproductive barrier that keeps the gene pools of these two species separate. Since the species can still interbreed when this prezygotic barrier is breached in the laboratory, the genetic divergence between the species is likely to be small. This suggests that speciation in nature has occurred relatively recently.

Results

Under normal light, females of each species strongly preferred males of their own species. But under orange light, females of each species responded indiscriminately to males of both species. The resulting hybrids were viable and fertile.

Conclusion

The researchers concluded that mate choice by females based on male breeding coloration can act as a reproductive barrier that keeps the gene pools of these two species separate. Since the species can still interbreed when this prezygotic barrier is breached in the laboratory, the genetic divergence between the species is likely to be small. This suggests that speciation in nature has occurred relatively recently.

Data from O. Seehausen and J. J. M. van Alphen, The effect of male coloration on female mate choice in closely related Lake Victoria cichlids (*Haplochromis nyererei* complex), *Behavioral Ecology and Sociobiology* 42:1–8 (1998).

WHAT IF? *Suppose that female cichlids living in the murky waters of a polluted lake could not distinguish colors well. In such waters, how might the gene pools of these species change over time?*

*For suggested answer, see*Appendix A*.*

Habitat Differentiation

Sympatric speciation can also occur when a subpopulation exploits a habitat or resource not used by the parent population. Consider the North American apple maggot fly (*Rhagoletis pomonella*), a pest of apples. The fly's original habitat was the native hawthorn tree (see Figure 24.3a), but about 200 years ago, some populations colonized apple trees that had been introduced by European settlers. Apple maggot flies usually mate on or near their host plant. This results in a prezygotic barrier (habitat isolation) between populations that feed on apples and populations that feed on hawthorns. Furthermore, because apples mature more quickly than hawthorn fruit, natural selection has favored apple-feeding flies with rapid development. These apple-feeding populations now show temporal isolation from the hawthorn-feeding *R. pomonella*, providing a second prezygotic barrier to gene flow between the two populations. Researchers also have identified alleles that benefit the flies that use one host plant but harm the flies that use the other host plant. Natural selection operating on these alleles has provided a postzygotic barrier to reproduction, further limiting gene flow. Altogether, although the two populations are still classified as subspecies rather than separate species, sympatric speciation appears to be well under way.

Allopatric and Sympatric Speciation: *A Review*

Now let's recap the processes by which new species form. In allopatric speciation, a new species forms in geographic isolation from its parent population. Geographic isolation severely restricts gene flow. Intrinsic barriers to reproduction with members of the parent population may then arise as a by-product of genetic changes that occur within the isolated population. Many different processes can produce such genetic changes, including natural selection under different environmental conditions, genetic drift, and sexual

selection. Once formed, reproductive barriers that arise in allopatric populations can prevent interbreeding with the parent population even if the populations come back into contact.

Sympatric speciation, in contrast, requires the emergence of a reproductive barrier that isolates a subset of a population from the remainder of the population in the same area. Though rarer than allopatric speciation, sympatric speciation can occur when gene flow to and from the isolated subpopulation is blocked. This can occur as a result of polyploidy, a condition in which an organism has extra sets of chromosomes. Sympatric speciation also can result from sexual selection. Finally, sympatric speciation can occur when a subset of a population becomes reproductively isolated because of natural selection that results from a switch to a habitat or food source not used by the parent population.

Having reviewed the geographic context in which species originate, we'll next explore in more detail what can happen when new or partially formed species come into contact.

HHMI BioInteractive Video

Anole Lizards: An Example of Speciation

image

Play

image

Play

For more resources from HHMI BioInteractive, go to www.biointeractive.org.

Concept 24.3: Hybrid zones reveal factors that cause reproductive isolation

What happens if species with incomplete reproductive barriers come into contact with one another? One possible outcome is the formation of a **hybrid zone**, a region in which members of different species meet and mate, producing at least some offspring of mixed ancestry. In this section, we'll explore hybrid zones and what they reveal about factors that cause the evolution of reproductive isolation.

Patterns Within Hybrid Zones

Some hybrid zones form as narrow bands, such as the one depicted in **Figure 24.13** for the yellow-bellied toad (*Bombina variegata*) and its close relative, the fire-bellied toad (*B. bombina*). This hybrid zone, represented by the red line on the map, extends for 4,000 km but is less than 10 km wide in most places. The hybrid zone occurs where the higher-altitude habitat of the yellow-bellied toad meets the lowland habitat of the fire-bellied toad. Across a given "slice" of the zone, the frequency of alleles specific to yellow-bellied toads typically decreases from close to 100% at the edge where only yellow-bellied toads are found to around 50% in the central portion of the zone and to close to 0% at the edge where only fire-bellied toads are found.

Figure 24.13

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A narrow hybrid zone for *Bombina* toads in Europe.
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Hybrid zone for Bombina toads occur where the habitats of the two species meet. Graph of Frequency of B variegata-specific allele versus distance from hybrid zone center is shown.

Hybrid zone for Bombina toads occur where the habitats of the two species meet. Fire-bellied toad, Bombina bombina lives at lower altitudes. Yellow-bellied toad, Bombina variegata lives at higher altitudes. Graph of Frequency of B variegata-specific allele versus distance from hybrid zone center (kilometers) is shown. The scale on the x-axis, from left to right, is from negative 40 to 20. The scale on the y-axis is from zero to 1.0. The yellow-bellied toad range is from negative 40 to negative 5; the hybrid zone is from negative 4 to 5; and the fire-belied toad range is from 6 to 20. The frequency of B. variegata-specific allele falls from 1.0 at negative 40 to 0.90 at negative 3. It then drops to 0.05 at 2 and declines to 0.01 at 20.

The graph shows species-specific allele frequencies across the width of the zone near Krakow, Poland, averaged over six genetic loci. A value of 1.0 indicates that all individuals were yellow-bellied toads, 0 indicates that all individuals were fire-bellied toads, and intermediate frequencies indicate that some individuals were of mixed ancestry.

Does the graph indicate that gene flow is spreading fire-bellied toad alleles into the range of the yellow-bellied toad? Explain.

*For suggested answer, see*Appendix A.**

What causes such a pattern of allele frequencies across a hybrid zone? We can infer that there is an obstacle to gene flow—otherwise, alleles from one parent species would also be common in the gene pool of the other parent species. Are geographic barriers reducing gene flow? Not in this case, since the toads can move throughout the hybrid zone. A more important factor is that hybrid toads have increased rates of embryonic mortality and a variety of morphological abnormalities, including ribs that are fused to the spine and malformed tadpole mouthparts. Because the hybrids have poor survival and reproduction, they produce few

viable offspring with members of the parent species. As a result, hybrid individuals rarely serve as a stepping-stone from which alleles are passed from one species to the other. Outside the hybrid zone, additional obstacles to gene flow may be provided by natural selection in the different environments in which the parent species live.

Hybrid zones typically are located wherever the habitats of the interbreeding species meet. Those regions often resemble a group of isolated patches scattered across the landscape—more like the pattern of spots on a Dalmatian than the continuous band shown in Figure 24.13. But regardless of whether they have complex or simple spatial patterns, hybrid zones form when two species lacking complete barriers to reproduction come into contact. What happens when the habitats of the interbreeding species change over time?

Hybrid Zones and Environmental Change

A change in environmental conditions can alter where the habitats of interbreeding species meet. When this happens, an existing hybrid zone can move to a new location, or a novel hybrid zone may form.

For example, black-capped chickadees (*Poecile atricapillus*) and Carolina chickadees (*P. carolinensis*) interbreed in a narrow hybrid zone that runs from New Jersey to Kansas. Recent studies have shown that the location of this hybrid zone has shifted northward as the climate has warmed **(Figure 24.14)**. In another example, a series of warm winters prior to 2003 enabled the southern flying squirrel (*Glaucomys volans*) to expand northward into the range of the northern flying squirrel, *G. sabrinus*. Previously, the ranges of these two species had not overlapped. Genetic analyses showed that these flying squirrels began to hybridize where their ranges came into contact, thereby forming a novel hybrid zone induced by climate change.

Figure 24.14

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A shift in a hybrid zone resulting from climate change.
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Part a is a black-capped chickadee (Poecile atricapilius), and part b is a Carolina chickadee (Poecile carolinensis. The latter is darker than the former.

Black-capped **(a)** and Carolina chickadees **(b)** interbreed in a hybrid zone that runs from Kansas to New Jersey. In Pennsylvania, the center of this hybrid zone moved 12 km to the north from 2002 to 2012. This shift is consistent with predictions based on the warmer winter temperatures that have resulted from climate change.

Finally, note that a hybrid zone can be a source of novel genetic variation that improves the ability of one or both parent species to cope with changing environmental conditions. This can occur when an allele found

only in one parent species is transferred first to hybrid individuals and then to the other parent species when hybrids breed with the second parent species. Recent genetic analyses have shown that hybridization has been a source for such novel genetic variation in some insect, mammal, bird, and plant species. In the **Problem-Solving Exercise**, you can examine one such example: a case in which hybridization may have led to the transfer of insecticide-resistance alleles between species of mosquitoes that transmit malaria.

Problem-Solving Exercise

AP

Is hybridization promoting insecticide resistance in mosquitoes that transmit malaria?

Malaria is a leading cause of human illness and mortality worldwide, with 200 million people infected and 600,000 deaths each year. In the 1960s, the incidence of malaria was reduced, owing to the use of insecticides that killed mosquitoes in the genus *Anopheles*, which transmit the disease from person to person. But today, mosquitoes are becoming resistant to insecticides—causing a resurgence in malaria.

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Insecticide-treated bed nets have helped reduce cases of malaria in many countries, but resistance to insecticides is rising in mosquito populations.

In this exercise, you will investigate whether alleles encoding resistance to insecticides have been transferred between closely related species of *Anopheles*.

Your Approach The principle guiding your investigation is that DNA analyses can detect the transfer of resistance alleles between closely related mosquito species. To find out whether such transfers have occurred, you will analyze DNA results from two species of mosquitoes that transmit malaria (*Anopheles gambiae* and *A. coluzzii*) and from A. gambiae×A. coluzzii hybrids.

Your Data Resistance to DDT and other insecticides in *Anopheles* is affected by a sodium channel gene, kdr. The r allele of this gene confers resistance, while the wild-type (+/+) genotype is not resistant. Researchers sequenced the kdr gene from mosquitoes collected in Mali during three time periods: pre-2006 (2002 and 2004), 2006, and post-2006 (2009–2012). *A. gambiae* and *A. coluzzii* were collected during all three time periods, but their hybrids only occurred in 2006, the first year that insecticide-treated bed nets were used to reduce the spread of malaria. A likely explanation is that the introduction of the treated bed nets may have briefly favored hybrid individuals, which are usually at a selective disadvantage.

A table for observed number of mosquitoes by kdr genotype is given.

Observed numbers of mosquitoes by k d r genotype
A. gambiae, Pre-2006
A. gambiae, 2006
A. gambiae, post-2006
Hybrids, 2006
A. coluzzii, pre-2006
A. coluzzii, 2006
A. coluzzii, post-2006

Your Analysis

-
  - Calculate the kdr genotype frequencies in *A. gambiae* for each time period. To do this, divide the number of individuals that have a given genotype by the total number of individuals observed for that time period.

- How did the kdr genotype frequencies change over time? Describe a hypothesis that accounts for these observations.
- How did the frequencies of kdr genotypes change over time in *A. coluzzii*? Describe a hypothesis that accounts for these observations.
- Do these results indicate that hybridization can lead to the transfer of adaptive alleles? Explain.
- Predict how the transfer of the r allele to *A. coluzzii* populations could affect the number of malaria cases in the years immediately following the transfer.

Instructors: A version of this Problem-Solving Exercise can be assigned in **Mastering Biology**.

Hybrid Zones over Time

Studying a hybrid zone is like observing a naturally occurring experiment on speciation. Will the hybrids become reproductively isolated from their parents and form a new species? This has occurred by polyploid speciation in some cases, as we saw for the formation of two new species of goatsbeard plants (see Figure 24.11). Hybridization also has led to what may turn out to be a new species of finch in the Galápagos archipelago: A 2018 study found that descendants of hybrids between the large cactus finch (*Geospiza conirostris*) and the medium ground finch (*G. fortis*) are becoming reproductively isolated from both parent species. As of 2019, this emerging species is known informally as "Big Bird."

In other cases, however, interspecific hybrids do not become reproductively isolated from their parent species. In such situations, there are three common outcomes for a hybrid zone over time: reinforcement of barriers, fusion of species, or **stability (Figure 24.15)**. Let's examine what studies suggest about these possibilities.

Figure 24.15

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    Formation of a hybrid zone and common outcomes for hybrids
over time.
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Three populations connected by gene flow. A barrier to gene flow is established and one population diverges. Gene flow is later re-established in the hybrid zone with various outcomes possible

The thick gray and purple arrows represent the passage of time.

- Three populations of a species are connected by gene flow.
- A barrier to gene flow is established.
- One population begins to diverge from the other two (gene flow remains in place only between the other two populations). Changing colors indicate that the isolated population changes over time.
- Later, gene flow is re-established in a hybrid zone. Hybrid individuals are present.
- Possible outcomes for hybrids:
- a. Reinforcement (strengthening of reproductive barriers, hybrids gradually cease to be formed).
- Or b. Fusion (weakening of reproductive barriers, the two species fuse).
- Or c. Stability (continued production of hybrid individuals).

WHAT IF? *Predict what might happen if gene flow were re-established at step 3 in this process.*

For suggested answer, see ***Appendix A***.

Reinforcement: Strengthening Reproductive Barriers

Hybrids often are less fit than members of their parent species. In such cases, natural selection should strengthen prezygotic barriers to reproduction, reducing the formation of unfit hybrids. Because this process involves *reinforcing* reproductive barriers, it is called **reinforcement**. If reinforcement is occurring, a logical prediction is that barriers to reproduction between species should be stronger for sympatric populations than for allopatric populations.

As an example, let's consider two species of European flycatcher, the pied flycatcher (*Ficedula hypoleuca*) and the collared flycatcher (*Ficedula albicollis*). In allopatric populations of these birds, males of the two species closely resemble one another, while in sympatric populations, the males look very different. Female flycatchers do not select males of the other species when given a choice between males from sympatric populations, but they frequently do make mistakes when selecting between males from allopatric populations. Thus, barriers to reproduction are stronger in birds from sympatric populations than in birds from allopatric populations, as you would predict if reinforcement were occurring. Similar results have been observed in a number of organisms, including some fishes, insects, plants, and other birds.

Fusion: Weakening Reproductive Barriers

Barriers to reproduction may be weak when two species meet in a hybrid zone. Indeed, so much gene flow may occur that reproductive barriers weaken further and the gene pools of the two species become increasingly alike. In effect, the speciation process reverses, eventually causing the two hybridizing species to fuse into a single species.

For example, genetic and morphological evidence indicates that the recent loss of the large tree finch from the Galápagos island of Floreana resulted from extensive hybridization with another finch species on that island. Such a situation also may be occurring among Lake Victoria cichlids **(Figure 24.16)**. Many pairs of ecologically similar cichlid species are reproductively isolated because the females of one species prefer to mate with males of one color, while females of the other species prefer to mate with males of a different color (see Figure 24.12). Results from field and laboratory studies indicate that murky waters caused by pollution have reduced the ability of females to use color to distinguish males of their own species from males of closely related species. In some polluted waters, many hybrids have been produced, leading to fusion of the parent species' gene pools and a loss of species.

Figure 24.16

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Fusion: the breakdown of reproductive barriers.
```

Pundamilia nyererei and Pundamilia pundamilia produce hybrid offspring Pundamilia turbid water in a location with turbid water.

Increasingly cloudy water in Lake Victoria over the past several decades may have weakened reproductive barriers between P. nyererei and P. pundamilia. In areas of cloudy water, the two species have hybridized extensively, causing their gene pools to fuse.

Stability: Continued Formation of Hybrid Individuals

Many hybrid zones are stable in the sense that hybrids continue to be produced. In some cases, this occurs because the hybrids survive or reproduce better than members of either parent species, at least in certain habitats or years. But stable hybrid zones have also been observed in cases where the hybrids are selected *against*—an unexpected result.

For example, hybrids continue to form in the *Bombina* hybrid zone even though they are strongly selected against. One explanation relates to the narrowness of the *Bombina* hybrid zone (see Figure 24.13). Evidence suggests that members of both parent species migrate into the zone from the parent populations located outside the zone, thus leading to the continued production of hybrids. If the hybrid zone were wider, this would be less likely to occur, since the center of the zone would receive little gene flow from distant parent populations located outside the hybrid zone.

Sometimes the outcomes in hybrid zones match our predictions (European flycatchers and cichlid fishes), and sometimes they don't (*Bombina*). But whether our predictions are upheld or not, events in hybrid zones can shed light on how barriers to reproduction between closely related species change over time. In the next section, we'll examine how interactions between hybridizing species can also provide a glimpse into the speed and genetic control of speciation.

Concept 24.4: Speciation can occur rapidly or slowly and can result from changes in few or

many genes

Darwin faced many questions when he began to ponder that “mystery of mysteries”—speciation. He found answers to some of those questions when he realized that evolution by natural selection helps explain both the diversity of life and the adaptations of organisms (see Concept 22.2). But biologists since Darwin have continued to ask fundamental questions about speciation. How long does it take for new species to form? And how many genes change when one species splits into two? Answers to these questions are also emerging.

The Time Course of Speciation

We can gather information about how long it takes new species to form from broad patterns in the fossil record and from studies that use morphological data (including fossils) or molecular data to assess the time interval between speciation events in particular groups of organisms.

Patterns in the Fossil Record

The fossil record includes many episodes in which new species appear suddenly in a geologic stratum, persist essentially unchanged through several strata, and then disappear. For example, there are dozens of species of marine invertebrates that make their debut in the fossil record with novel morphologies, but then change little for millions of years before becoming extinct. The term **punctuated equilibria** is used to describe these periods of apparent stasis punctuated by sudden change **(Figure 24.17a)**. Other species do not show a punctuated pattern; instead, they appear to have changed more gradually over long periods of time **(Figure 24.17b)**.

Figure 24.17

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Two models for the tempo of speciation.
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Compares punctuated model (new species change most as they branch from a parent species) and gradual model (species change slowly and steadily over time).

- Punctuated model. New species change most as they branch from a parent species and then change little for the rest of their existence. Punctuated model illustrated with a brown-orange parent moth that branches to form a brown-white moth species and an orange moth species. These species remain constant over time.

- Gradual model. Species diverge from one another more slowly and steadily over time. Gradual model illustrated with a brown-orange parent moth branching to create two new species of moths: a slightly lighter brown-orange species and a slightly darker brown-orange species. Over time, the lighter brown-orange species becomes brown-white and the darker brown-orange species becomes orange.

What might punctuated and gradual patterns tell us about how long it takes new species to form? Suppose that a species survived for 5 million years, but most of the morphological changes that caused it to be designated a new species occurred during its first 50,000 years—just 1% of the total existence of the species. Time periods this short (in geologic terms) often cannot be distinguished in fossil strata, in part because the rate of sediment accumulation may be too slow to separate layers this close in time. Thus, based on its fossils, the species would seem to have appeared suddenly and then lingered with little or no change before becoming extinct. Even though such a species may have originated more slowly than its fossils suggest (in this case taking up to 50,000 years), a punctuated pattern indicates that speciation occurred relatively rapidly. For species whose fossils changed much more gradually, we also cannot tell exactly when a new biological species formed, since information about reproductive isolation does not fossilize. However, it is likely that speciation in such groups occurred relatively slowly, perhaps taking millions of years.

Interview

Interview with Stephen Jay Gould: An “architect” of the concept of punctuated equilibria

Jay Gould.

Speciation Rates

The existence of fossils that display a punctuated pattern suggests that once the process of speciation begins, it can be completed relatively rapidly—a suggestion supported by many studies. For example, rapid speciation appears to have produced the wild sunflower *Helianthus anomalus*. Genetic evidence indicates that this species originated by the hybridization of two other sunflower species, *H. annuus* and *H. petiolaris*. The hybrid species *H. anomalus* is ecologically distinct and reproductively isolated from both parent species **(Figure 24.18)**. Unlike the outcome of allopolyploid speciation, in which there is a change in chromosome number after hybridization, in these sunflowers the two parent species and the hybrid all have the same number of chromosomes (2n=34). How, then, did speciation occur? To study this question, researchers performed an experiment designed to mimic events in nature **(Figure 24.19)**. Their results indicated that natural selection could produce extensive genetic changes in hybrid populations over short periods of time.

These changes appear to have caused the hybrids to diverge reproductively from their parents and form a new species, *H. anomalus*.

Figure 24.18

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A hybrid sunflower species and its dry sand dune habitat.
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Photo of plant with bright yellow flowers growing in arid sandy environment.

The wild sunflower Helianthus anomalus shown here originated via the hybridization of two other sunflowers, H. annuus and H. petiolaris, which live in nearby but moister environments.

Figure 24.19

Inquiry: How does hybridization lead to speciation in sunflowers?

Experiment

Loren Rieseberg and his colleagues crossed the two parent sunflower species, *H. annuus* and *H. petiolaris*, to produce experimental hybrids in the laboratory (for each gamete, only two of the n=17 chromosomes are shown).

A diagram of an experiment. Gametes from sunflower species H annuus and H petiolarus are crossed to produce F 1 experimental hybrid. 4 of the 2 n = 34 chromosomes are shown.

Note that in the first F1 generation, each chromosome of the experimental hybrids consisted entirely of DNA from one or the other parent species. The researchers then tested whether the F1 and subsequent generations of experimental hybrids were fertile. They also used species-specific genetic markers to compare the chromosomes in the experimental hybrids with the chromosomes in the naturally occurring hybrid *H. anomalus*.

Results

Although only 5% of the F1 experimental hybrids were fertile, after just four more generations the hybrid fertility rose to more than 90%. The chromosomes of individuals from this fifth hybrid generation differed

from those in the F1 generation (see above) but were similar to those in *H. anomalus* individuals from natural populations:

A diagram of the experiment results.

Chromosomes 1 and 2 are diagrammed and regions compared between H. anomalus and experimental hybrid.

- Chromosome 1: All 8 comparison regions match for H. anomalus chromosome and experimental hybrid chromosome.
- Chromosome 2: The experimental hybrid chromosome differs from the H. anomalus chromosome in 3 out of 10 comparison regions. The other regions are the same.

Conclusion

Over time, the chromosomes in the population of experimental hybrids became similar to the chromosomes of *H. anomalus* individuals from natural populations. This suggests that the observed rise in the fertility of the experimental hybrids may have occurred as selection eliminated regions of DNA from the parent species that were not compatible with one another. Overall, it appeared that the initial steps of the speciation process occurred rapidly and could be mimicked in a laboratory experiment.

Data from L. H. Rieseberg et al., Role of gene interactions in hybrid speciation: evidence from ancient and experimental hybrids, *Science* 272:741–745 (1996).

WHAT IF? *The increased fertility of the experimental hybrids could have resulted from natural selection for thriving under laboratory conditions. Evaluate this alternative explanation for the result.*

*For suggested answer, see*Appendix A.**

The sunflower example, along with the apple maggot fly, Lake Victoria cichlid, and fruit fly examples discussed earlier, suggests that new species can arise rapidly *once divergence begins*. But what is the total length of time between speciation events? This interval consists of the time that elapses before populations of a newly formed species start to diverge from one another plus the time it takes for speciation to be complete once divergence begins. It turns out that the total time between speciation events varies considerably. In a survey of data from 84 groups of plants and animals, speciation intervals ranged from 4,000 years (in cichlids of Lake Nabugabo, Uganda) to 40 million years (in some beetles). Overall, the time between speciation events averaged 6.5 million years and was rarely less than 500,000 years.

These data suggest that on average, millions of years may pass before a newly formed plant or animal species will itself give rise to another new species. As you'll read in Concept 25.4, this finding has implications for

how long it takes life on Earth to recover from mass extinction events. Moreover, the extreme variability in the time it takes new species to form indicates that organisms do not have an internal "speciation clock" that causes them to produce new species at regular intervals. Instead, speciation begins only after gene flow between populations is interrupted, perhaps by changing environmental conditions or by unpredictable events, such as a storm that transports a few individuals to a new area. Furthermore, once gene flow is interrupted, the populations must diverge genetically to such an extent that they become reproductively isolated—all before other events cause gene flow to resume, possibly reversing the speciation process (see Figure 24.16).

Studying the Genetics of Speciation

Studies of ongoing speciation (as in hybrid zones) can reveal traits that cause reproductive isolation. By identifying the genes that control those traits, scientists can explore a fundamental question of evolutionary biology: How many genes influence the formation of new species?

In some cases, the evolution of reproductive isolation results from the effects of a single gene. For example, in Japanese snails of the genus *Euhadra*, a change in a single gene results in a mechanical barrier to reproduction. This gene controls the direction in which the shells spiral. When their shells spiral in different directions, the snails' genitalia are oriented in a manner that prevents mating (Figure 24.3f and g show a similar example). Recent genetic analyses have uncovered other single genes that cause reproductive isolation in fruit flies or mice.

A major barrier to reproduction between two closely related species of monkey flower, *Mimulus cardinalis* and *M. lewisii*, also appears to be influenced by a relatively small number of genes. These two species are isolated by several prezygotic and postzygotic barriers. Of these, one prezygotic barrier, pollinator choice, accounts for most of the isolation: In a hybrid zone between *M. cardinalis* and *M. lewisii*, nearly 98% of pollinator visits were restricted to one species or the other.

The two monkey flower species are visited by different pollinators: Hummingbirds prefer the red-flowered *M. cardinalis*, and bumblebees prefer the pink-flowered *M. lewisii*. Pollinator choice is affected by at least two loci in the monkey flowers, one of which, the "yellow upper," or *yup*, locus, influences flower color **(Figure 24.20)**. By crossing the two parent species to produce F1 hybrids and then performing repeated backcrosses of these F1 hybrids to each parent species, researchers succeeded in transferring the *M. cardinalis* allele at this locus into *M. lewisii*, and vice versa. In a field experiment, *M. lewisii* plants with the *M. cardinalis yup* allele received 68-fold more visits from hummingbirds than did wild-type *M. lewisii*. Similarly, *M. cardinalis* plants with the *M. lewisii yup* allele received 74-fold more visits from bumblebees

than did wild-type *M. cardinalis*. Thus, a mutation at a single locus can influence pollinator preference and hence contribute to reproductive isolation in monkey flowers.

Figure 24.20

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A locus that influences pollinator choice.
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Four flowers are given.

Pollinator preferences provide a strong barrier to reproduction between Mimulus lewisii and M. cardinalis. After transferring the M. lewisii allele for a flower-color locus into M. cardinalis and vice versa, researchers observed a shift in some pollinators' preferences.

- Part a, Typical Mimulus lewisii (pink flower).
- Part b, M lewisii with an M cardinalis flower-color allele (red-orange flower).
- Part c, Typical Mimulus cardinals (red flower).
- Part d, M cardinalis with an M lewisii flower-colored allele (dark pink flower).

WHAT IF? *If* M. cardinalis *individuals that had the* M. lewisii yup *allele were planted in an area that housed both monkey flower species, how might the production of hybrid offspring be affected?*

*For suggested answer, see*Appendix A.*

In other organisms, the speciation process is influenced by larger numbers of genes and gene interactions. For example, hybrid sterility between two subspecies of the fruit fly *Drosophila pseudoobscura* results from gene interactions among at least four loci, and postzygotic isolation in the sunflower hybrid zone discussed earlier is influenced by at least 26 chromosome segments (and an unknown number of genes). Overall, studies suggest that few or many genes can influence the evolution of reproductive isolation and hence the emergence of a new species.

From Speciation to Macroevolution

As you've seen, speciation may begin with differences as small as the color on a cichlid's back. However, as speciation occurs again and again, such differences can accumulate and become more pronounced, eventually leading to the formation of new groups of organisms that differ greatly from their ancestors (as in the origin of whales from terrestrial mammals; see Figure 22.20). Moreover, as one group of organisms increases in size by producing many new species, another group of organisms may shrink, losing species to extinction. The cumulative effects of many such speciation and extinction events have helped shape the

sweeping evolutionary changes that are documented in the fossil record. In the next chapter, we turn to such large-scale evolutionary changes as we begin our study of macroevolution.